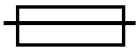
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

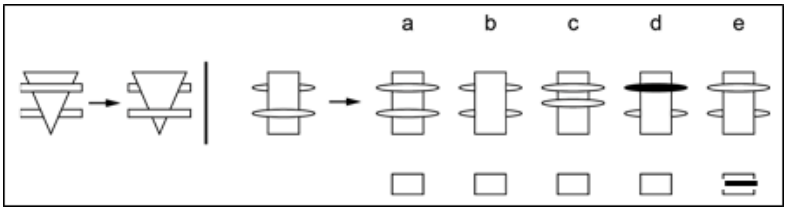
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 20 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 33 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



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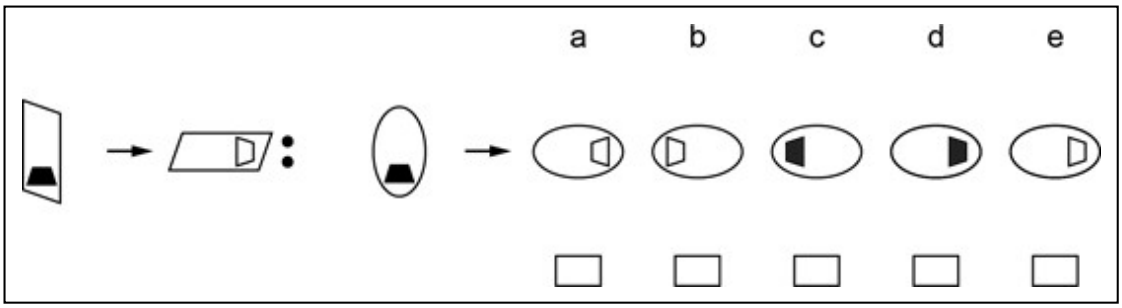

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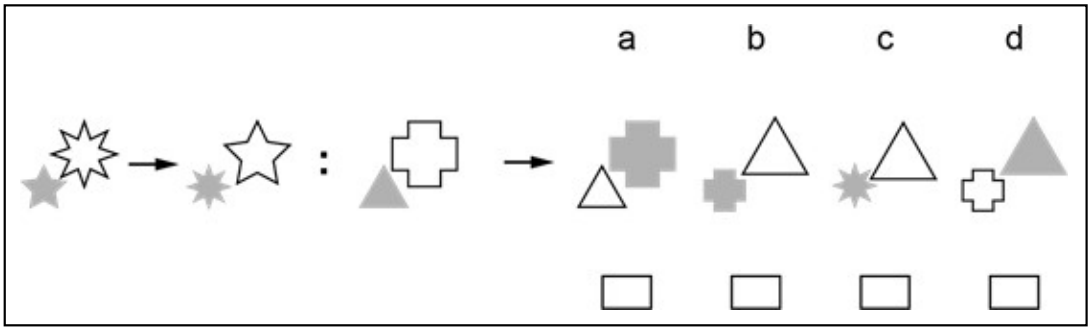

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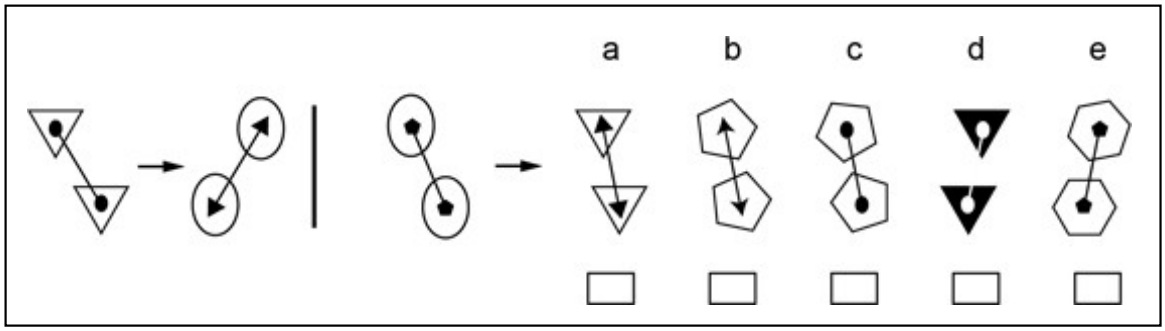
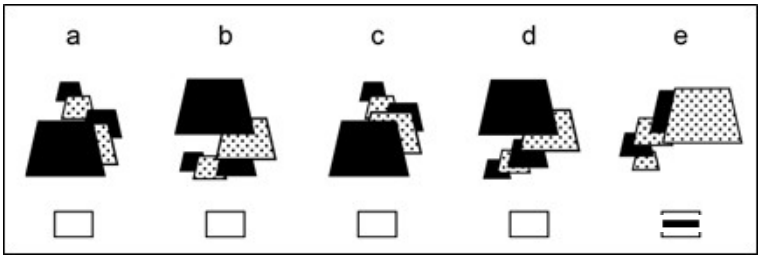


2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



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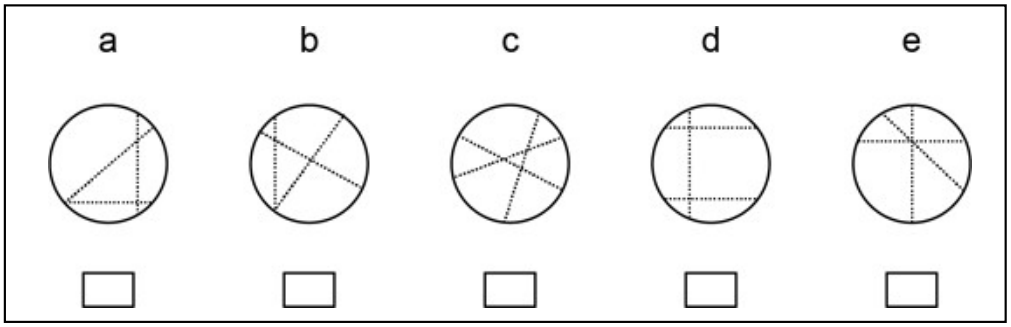

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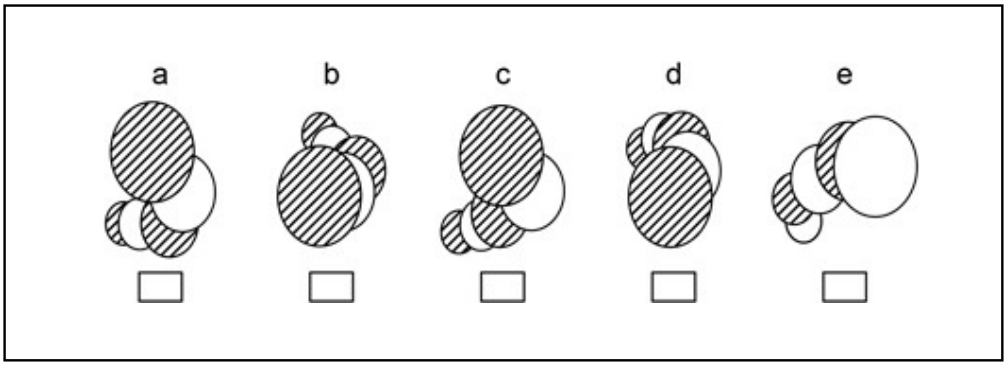

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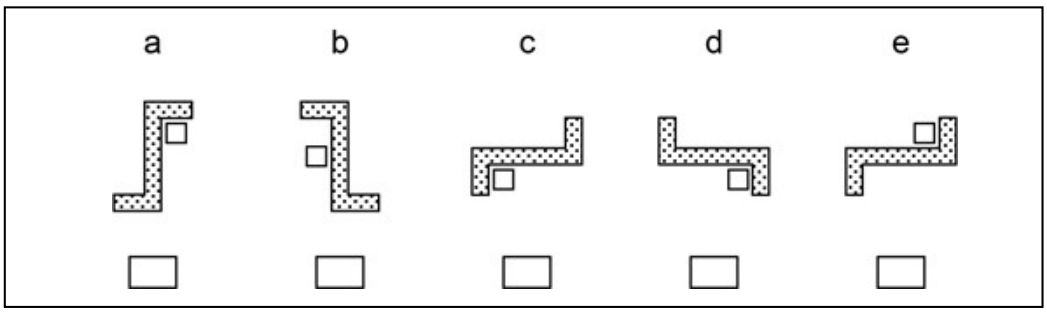
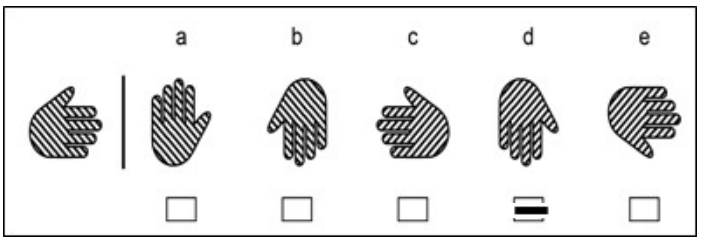


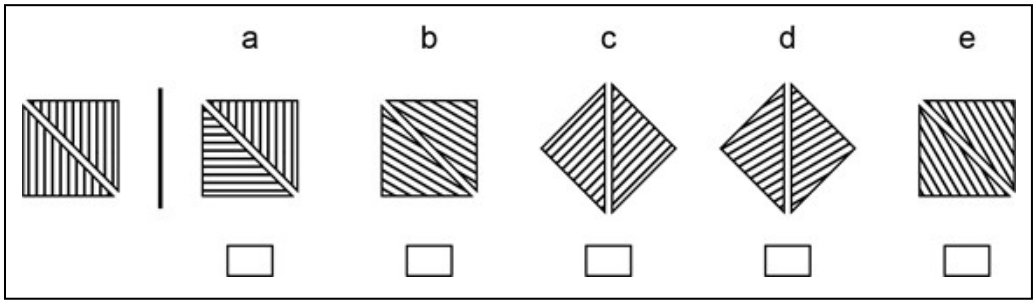
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



1



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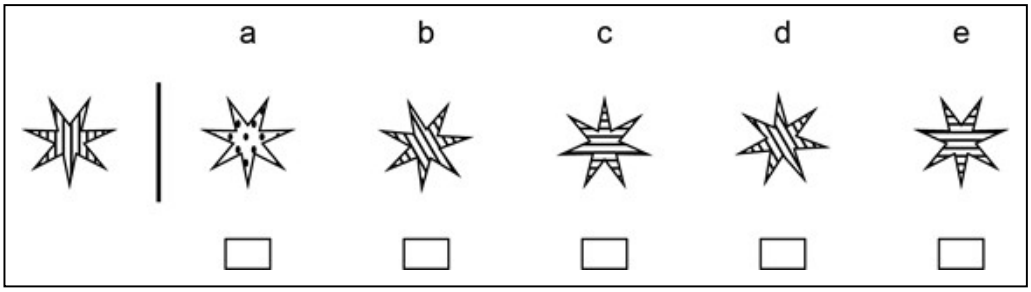

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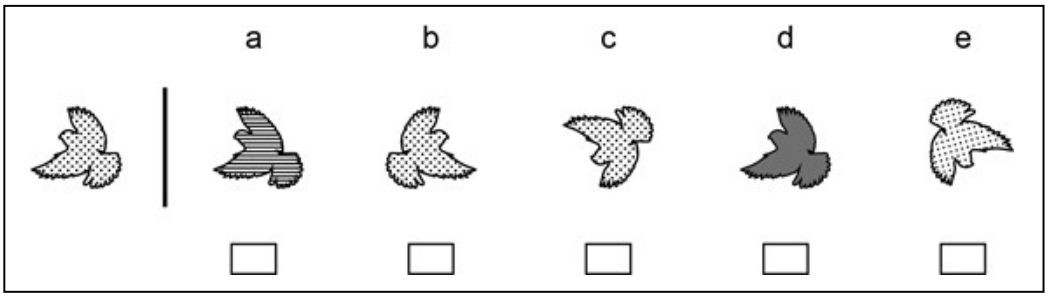
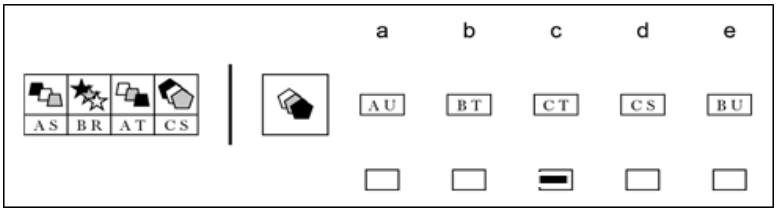


4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

Example



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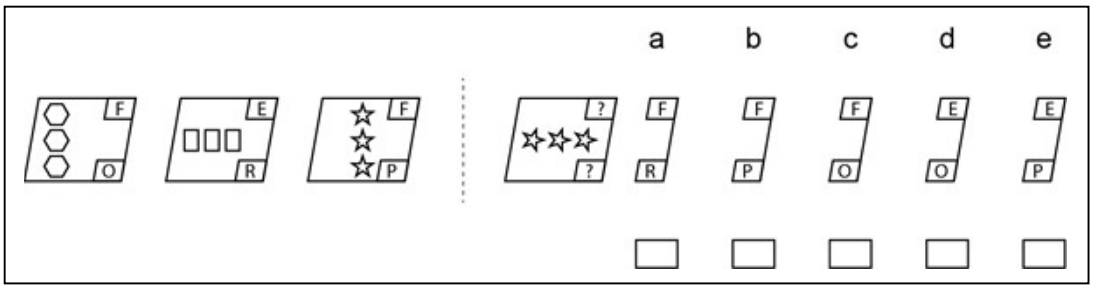

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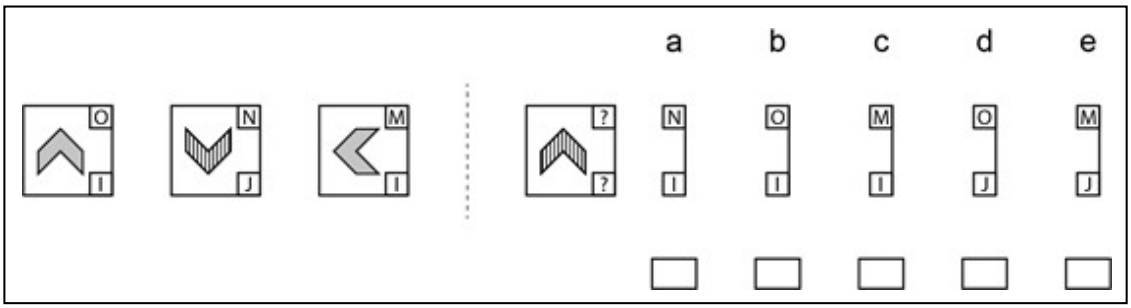

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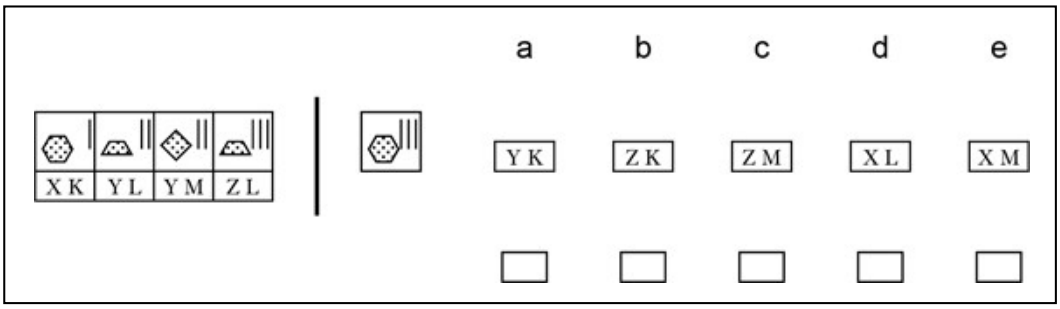
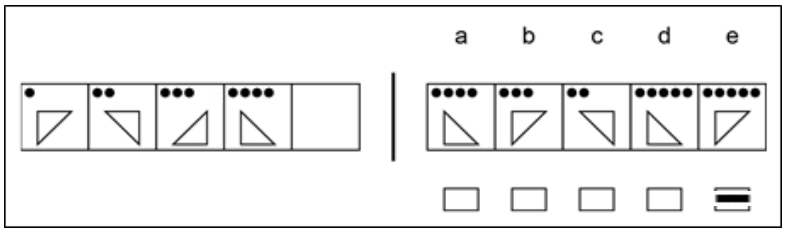


5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



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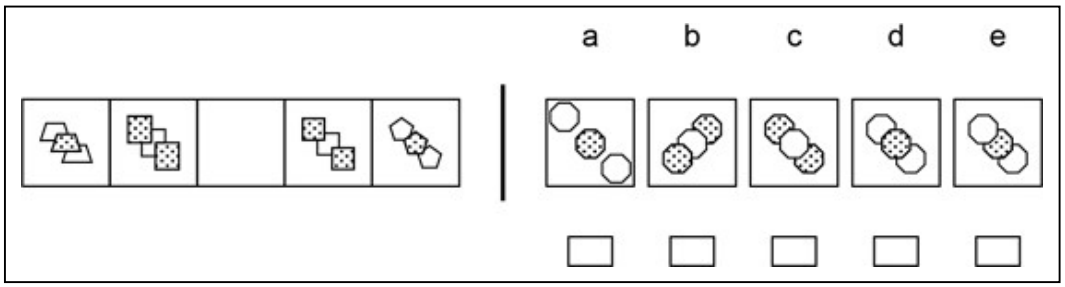

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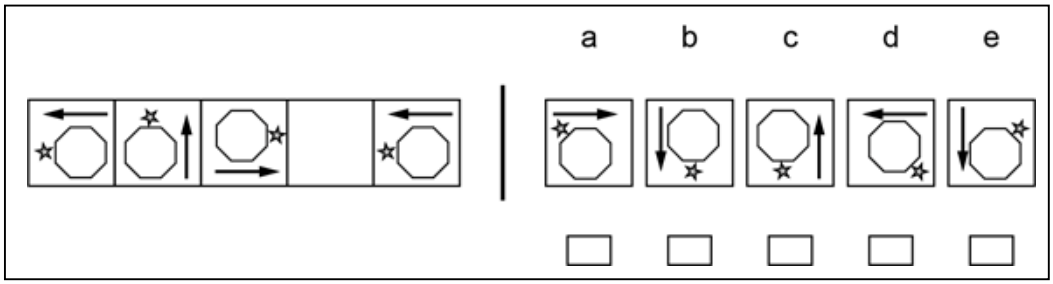

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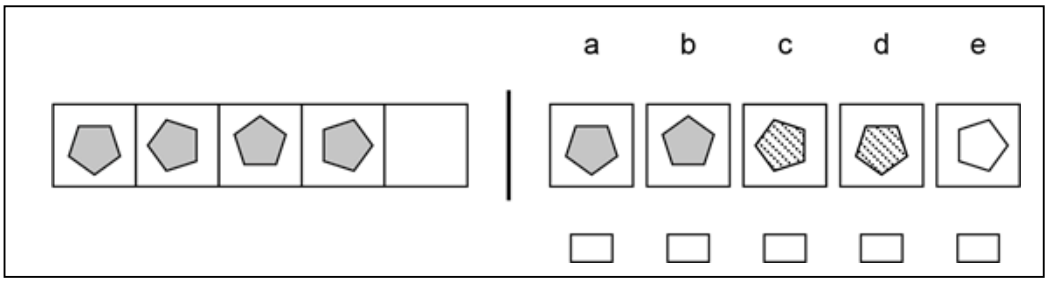
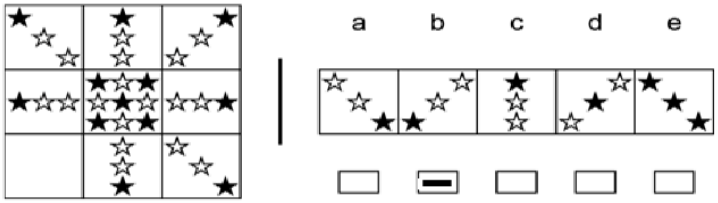


6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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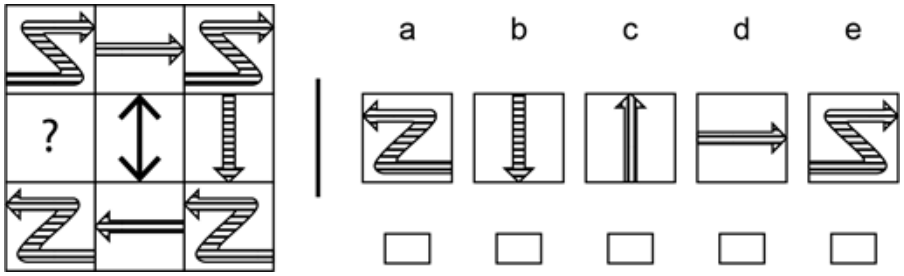

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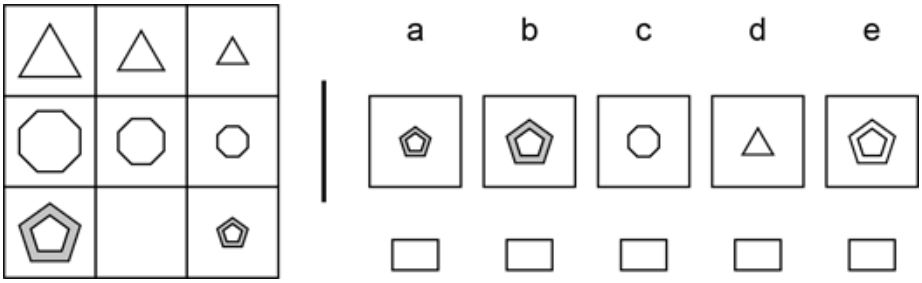

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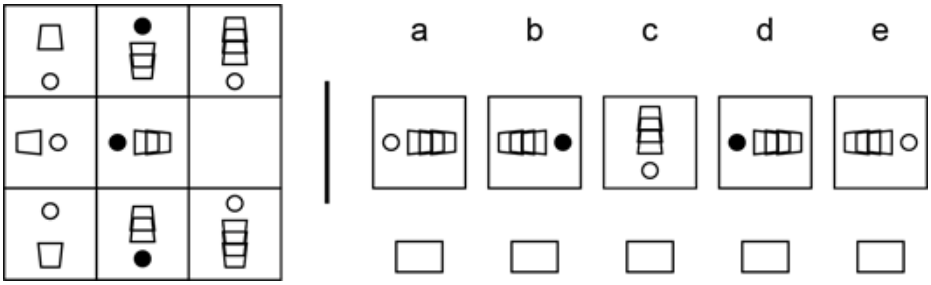
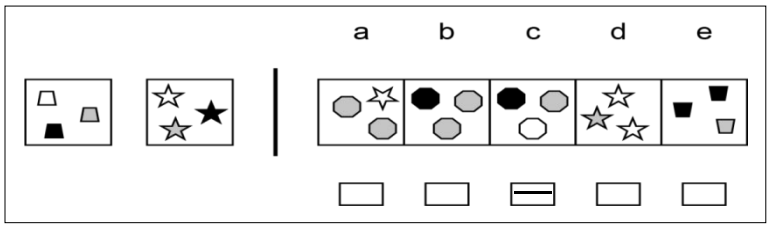


7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



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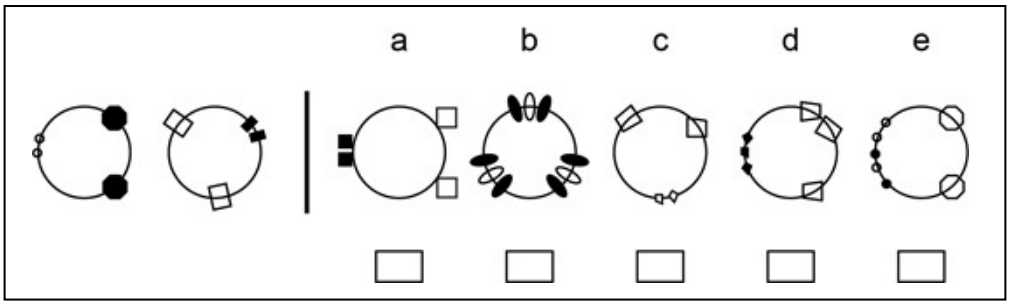

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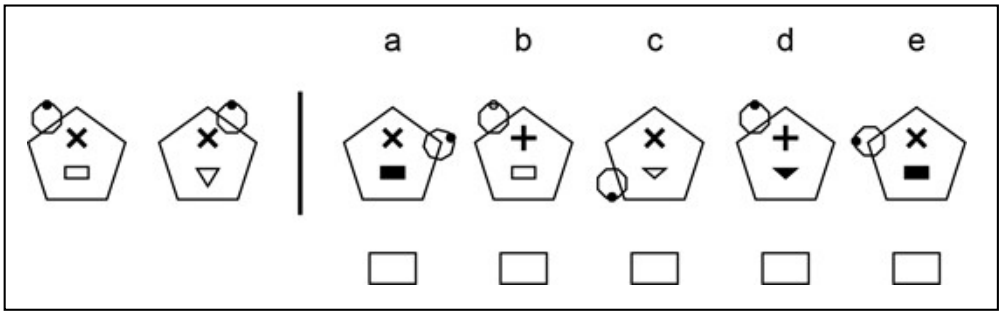

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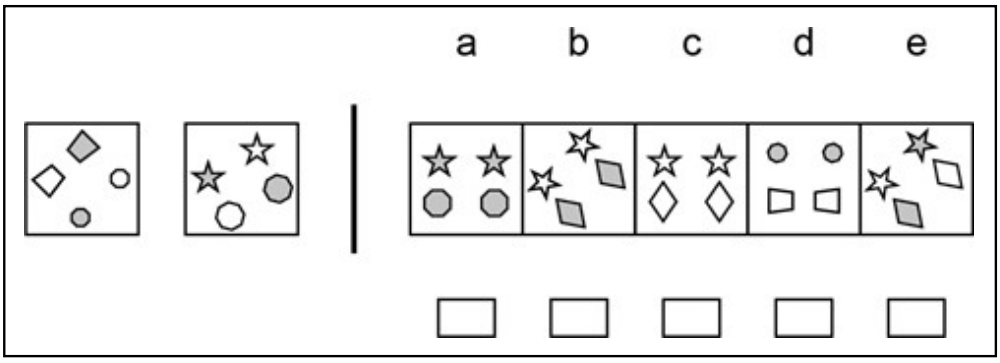
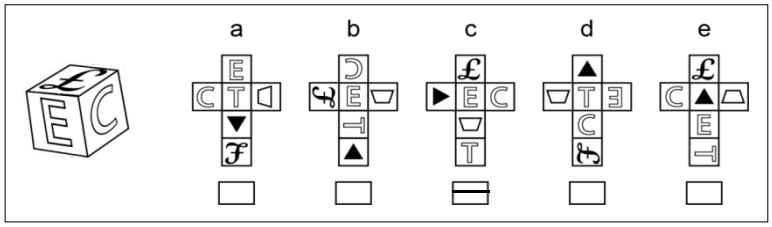


8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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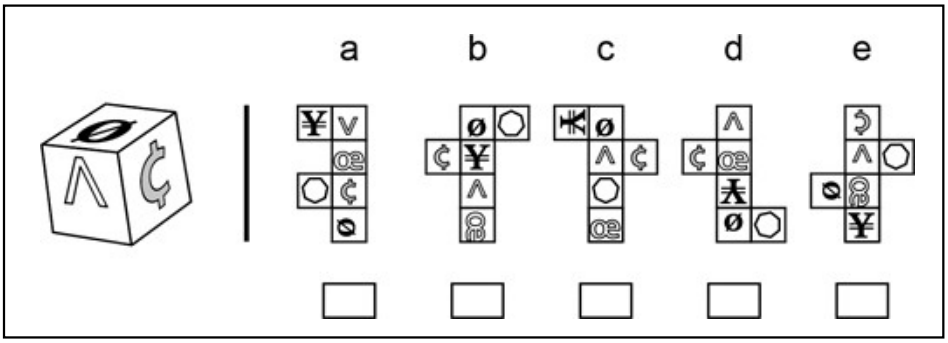

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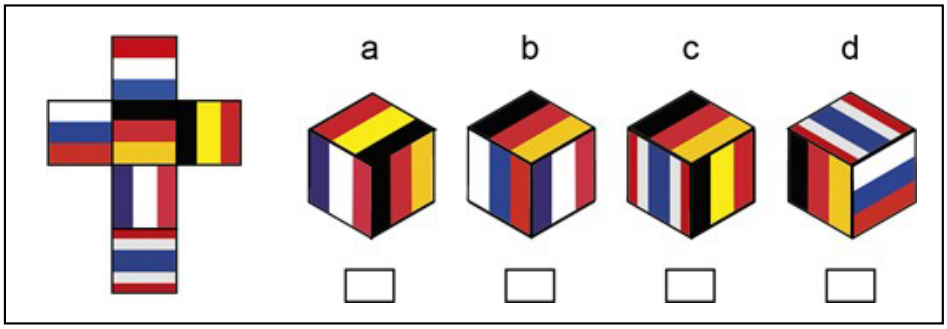

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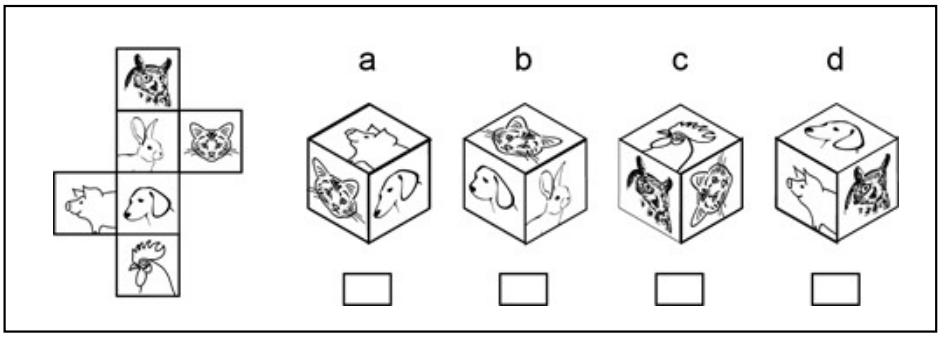
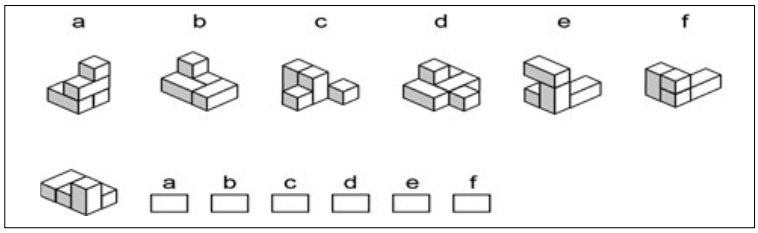


9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

Example



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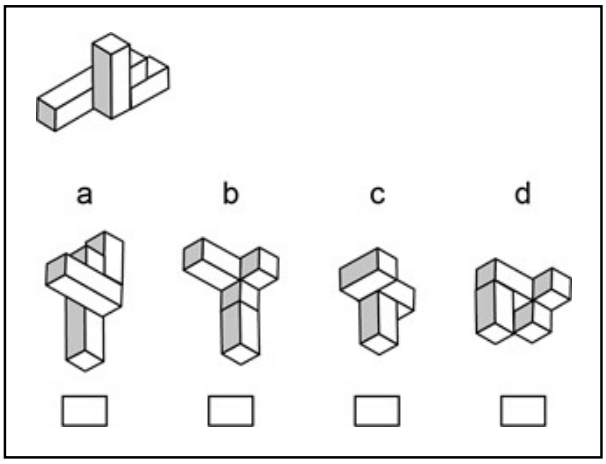

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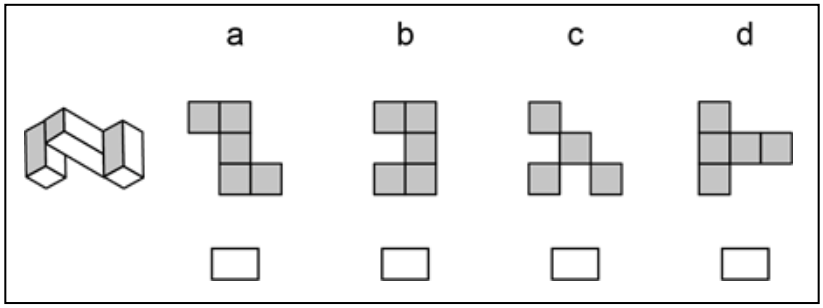

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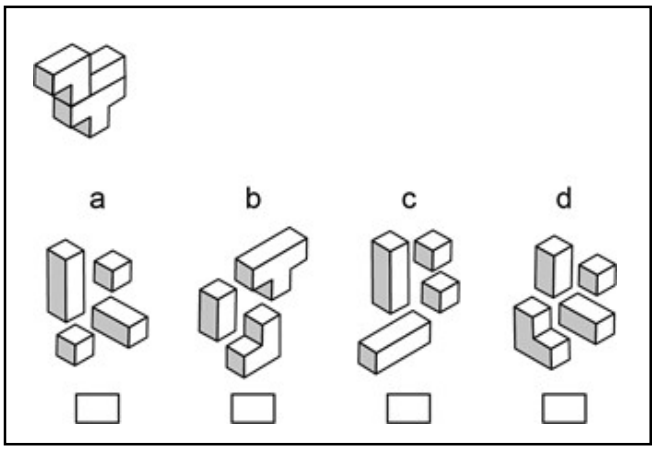
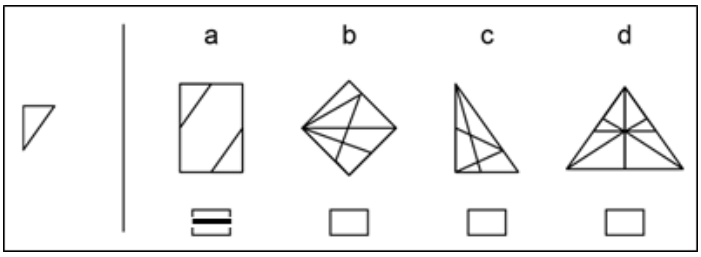


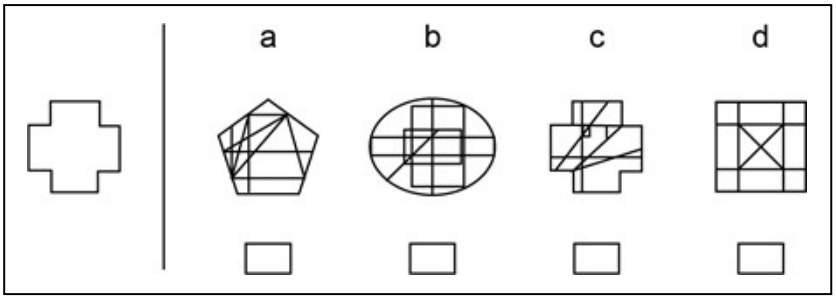
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

Example



1



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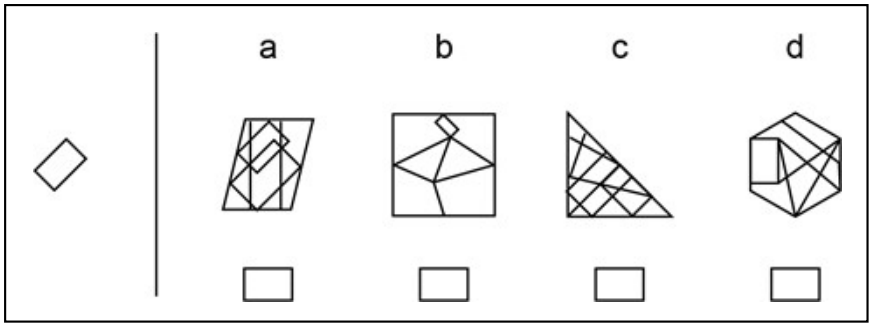

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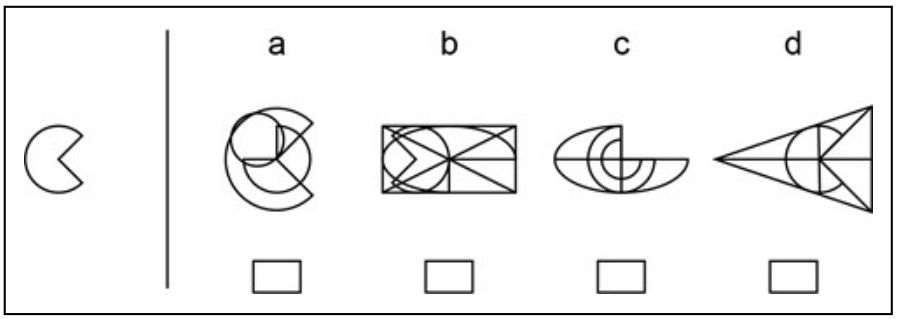
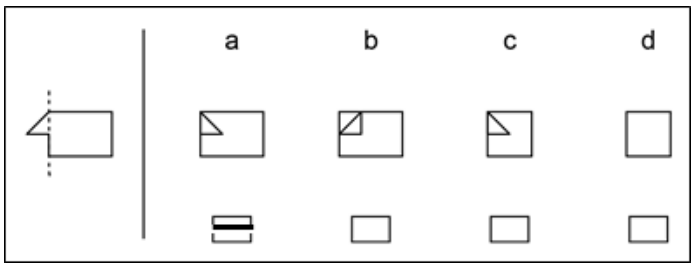


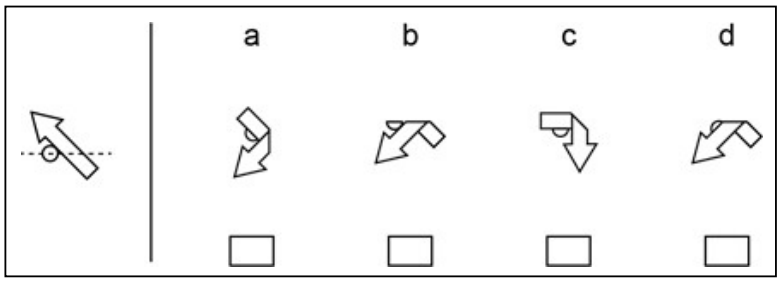
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

Example



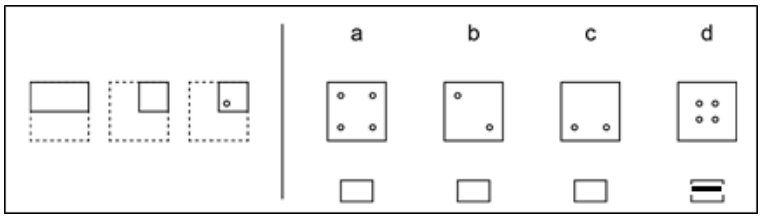
1



12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

Example



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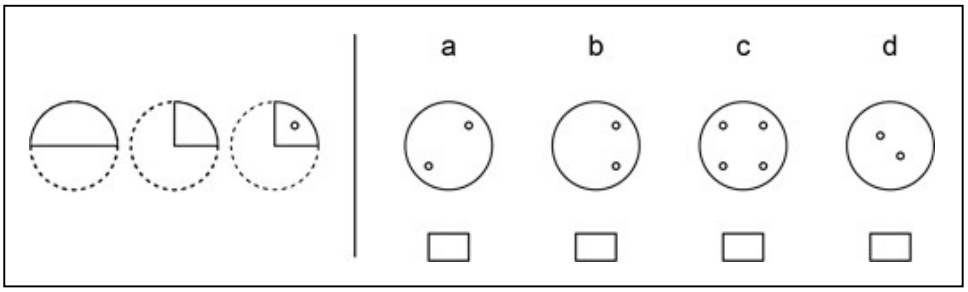
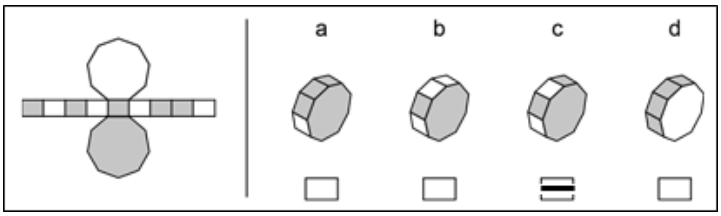


13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

Example



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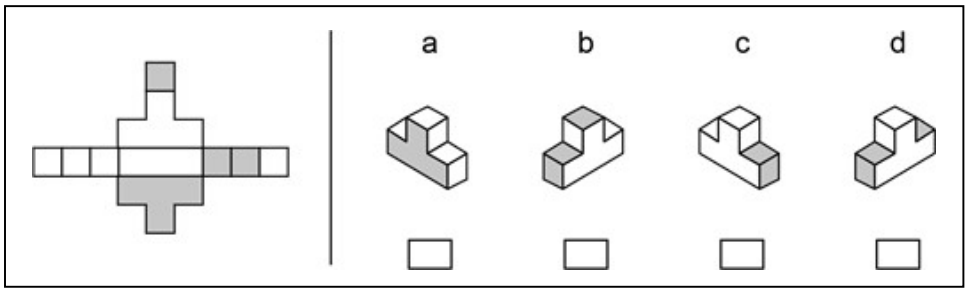


Non-Verbal Reasoning

Like Shapes

1. **E** - (The outer shape rotates 90 degrees anti-clockwise, with the inner shape remaining in the same respective position within the outer shape. The inner shape rotates 180 degrees and also changes from black to white.)
2. **B** - (The grey shape and white shape swap positions, size and colour.)
3. **C** - (The inside black shape that is attached to the line becomes the large outer white shape and the outer shape shrinks and becomes the small black shape attached to the line inside.)

Odd One Out

1. **E** - (In all the other answer options the dotted lines intersect at more than one point.)
2. **E** - (Answers A, B, C and D all start with a shape with diagonal lines on top then alternate between white and diagonal lines. However, E starts with white on top then alternates between diagonal lines and white and so is the odd one out.)
3. **B** - (All of the answers except B have the square in a corner, or within a right angle, but B does not so is the odd one out.)

Rotations

1. **C** - (C is a rotation of around 45 degrees clockwise of the original.)
2. **D** - (D is a rotation of around 45 degrees anti clockwise of the original.)
3. **E** - (E is a rotation of around 225 degrees clockwise of the original.)

Code Breaker

1. **E** - (You can work out that F stands for the three shapes being vertical rather than horizontal as there are two of them. Therefore, this means that E stands for the shapes being horizontal. All of the bottom letters are different, and the only difference between each of them is the type of the three sets of shapes. The star is therefore P. So, the final answer is EP.)
2. **D** - (You can work out that I stands for the chevron being lightly shaded as there are two of them. Therefore, J stands for the chevron having lines through it. All of the top letters are different and the only difference

between the three squares is that the chevrons are all pointing in different directions. The chevron pointing upwards is therefore O. So, the final answer is OJ.)

3. **B** - (You can work out that Y stands for there being two vertical lines as there are two of them. Therefore, X stands for there being one vertical line and Z stands for there being three vertical lines. In the right hand letters there are two Ls and the similarity between the two is that both of the shapes are trapeziums. So, K must be the shape being a hexagon. So, the final answer is ZK.)

Missing Sequence

1. **D** - (As you move from left to right the order of the colour of the shapes alternates between white, dotted, white and dotted, white, dotted and also the middle shape alternates between being in front of the other two shapes and being behind the other two shapes. The missing square should be white, dotted, white and should have the middle square in front of the other two, so the answer must be D.)
2. **B** - (As you move from left to right there are two changes in each square; firstly, the arrow rotates around the square anti-clockwise, moving one side at a time, and secondly, the star moves 90 degrees clockwise around the octagon. The missing square should have the arrow at the top of the square and the star on the right hand side of the octagon and so the answer is A.)
3. **A** - (As you move left to right the pentagons simply rotate 90 degrees clockwise. The missing square should be a rotation 90 degrees clockwise so must be A.)

Missing Square

1. **B** - (The squares in the bottom row are reflections through a vertical line of the squares in the top row. However when you move horizontally across a row, there is no change in the direction that the arrows are pointing in each square and they all point the same way. Therefore, in the middle row, the right hand square is pointing down so the left hand square must do the same. Also, the only square to point up is C, however the arrow in C has vertical lines whereas all the arrows should have horizontal lines. So the answer must be B.)

2. **B** - (As you can see from the top two rows, the shapes are decreasing in size as they move from left to right and that is the only change. A is the same size as the bottom right square so the answer must be B.)
3. **E** - (As you move across the two completed rows from left to right, a number of things happen: the small circle alternates between white and black, the number of trapeziums increases by two and then by one and the entire square is rotated 180 degrees. So the missing square should have a white circle, have four trapeziums and should be pointing to the left. The answer is therefore E.)

Most Like

1. **A** - (On one side on both of the shapes, there are two small shapes in a pair which then get bigger and either change from white to black or black to white on the opposite side of the circle. The only answer that matches this is A as the two small white circles turn to black and get larger on the opposite size. B, D and E all have too many shapes and in C there is one black and one white in each pair, rather than the same colour.)
2. **C** - (The shapes on the left both have one white heptagon and one small black pentagon on the outline of the shape and they also both have one black cross and one white shape inside, with the black cross being the top of the two shapes. The only shape with one white heptagon and one small black pentagon on the outline, one black cross and one white shape inside and with the black cross being the top of the two shapes is C.)
3. **E** - (In both squares on the left, the colours of the four shapes inside alternate between white and lightly shaded. The only answer where the colours of the four shapes follows this pattern is A.)

Nets

1. **C** - (You can see from the cube that the bottom of the ø sign lines up with the top of the ^ sign. The right hand side of the ^ sign also lines up with the left hand side of the C like symbol. From this information you can see that the answer is C.)
2. **B** - (You can see from the net that the middle blue part of the Russian flag meets the middle red part of the German flag. When formed in to a cube the bottom blue part of the Russian flag will meet the left blue part of the French flag.)

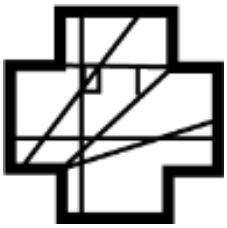
3. **C** - (You can see from the net that, when folded into a cube, the top of the cat's ears will be facing the owl. The chicken will then come all the way around so that the bottom of the chicken's neck is facing the owl's ears.)

3D Shapes

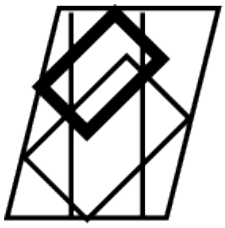
1. **A** - (The shape is rotated approximately 45 degrees anti-clockwise in the y-axis and then 90 degrees anti-clockwise in the z-axis.)
2. **A** - (Viewed from above, the shape will appear as a column of three squares length with one square coming off either end.)
3. **B** - (B must be the correct answer as it is the only option with a 'T' shaped block.)

Hidden Shape

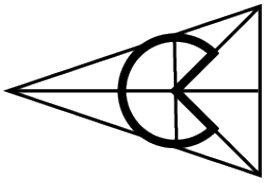
1. **C** -



2. **A** -



3. **D** -

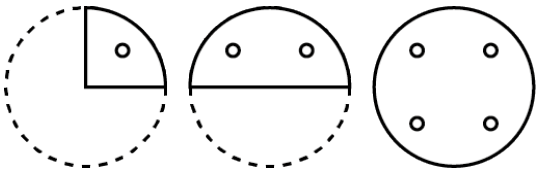


Fold Along the Dotted Line

1. **B** - (The top half of the shape folds over the bottom so that the arrow now points towards the bottom left corner and a semi-circle is visible next to the arrow.)

Hole Punch

1. ***C*** -



Shaded Nets

1. **B** - (When the net is folded up and viewed from the side of the white 'T' shape, the top square and two of the squares to the left will be shaded.)